

Lab 22 — Applying the CompTIA Six-Step Troubleshooting Methodology

Course: CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)

Topic 05: Hardware and Network Troubleshooting — Core 1, 28% of Core 1

Exam objective: Apply the CompTIA six-step troubleshooting methodology in order to a real fault, documenting each step (Core 1 objective 5.1).

Goal

The methodology is examinable in order and is the framework every other troubleshooting lab in this course hangs from. You apply all six steps to a fault, writing the evidence at each step, and produce the documentation that step six actually requires.

What you'll produce

A completed six-step troubleshooting record for one fault, with evidence and a preventive measure at each step.

Tools and equipment

Training PC with an injected fault, ticketing template, Killercoda Ubuntu Playground

Browser tools used in this lab

- **Killercoda Ubuntu Playground** — <https://killercoda.com/playgrounds/scenario/ubuntu>

Killercoda Ubuntu Playground — a real Linux machine in the browser

Terminal — root@controlplane

```
$ cat /etc/os-release | grep PRETTY
PRETTY_NAME="Ubuntu 22.04.3 LTS"
$ apt-get update -qq && apt-get install -y net-tools dnsutils
Reading package lists... Done
$ ip -brief addr show
lo        UNKNOWN  127.0.0.1/8
eth0      UP        10.244.0.14/24
$ dig +short www.tertiarycourses.com.sg
104.22.35.191
$ chmod 640 ~/applus/logs/system.log && ls -l ~/applus/logs/
-rw-r----- 1 root root 118 Aug 19 09:14 system.log
$ grep -c 'ERROR' ~/applus/logs/system.log
2
$ _
```

- 1 Open the URL and wait for the prompt. You get root on a real Ubuntu machine — nothing to install.
- 2 Start every lab with apt-get update, then install only the packages that lab needs.
- 3 The playground RESETS when idle. If your terminal disappears, reopen it and re-run the setup steps.

Killercoda Ubuntu Playground — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Write the six steps in order before you start, because the order itself is examinable: identify the problem; establish a theory of probable cause; test the theory; establish a plan of action and implement it; verify full functionality and implement preventive measures; document findings, actions and outcomes.
2. Step 1 — identify the problem. Interview the user with open questions: what exactly happens, when did it start, what changed, and can you reproduce it on demand.
3. Step 1 continued — always ask what changed, because most faults follow a recent change: a new driver, an update, a moved cable or new software.
4. Step 1 continued — back up user data before making any change, so that your troubleshooting

can never be the cause of data loss.

5. Step 2 — establish a theory of probable cause, questioning the obvious first and listing at least three candidate causes ranked by likelihood.
6. Step 3 — test the theory. Design a test that will disprove the theory if it is wrong, not merely one that confirms what you already believe.
7. Step 3 continued — if the theory is not confirmed, establish a new theory or escalate. Record that escalation is a legitimate step in the methodology, not a failure.
8. Step 4 — establish a plan of action, referring to vendor documentation, and state the rollback position before you implement anything.
9. Step 5 — verify full system functionality, testing not only the reported fault but the functions around it that your change might have affected.
10. Step 5 continued — implement preventive measures so the same fault does not recur, and state the specific measure you applied.
11. Step 6 — document findings, actions and outcomes in the ticket, in language a colleague could follow without speaking to you.
12. Practise the discipline on the KillerCoda playground by diagnosing a deliberately broken DNS configuration through all six steps and recording your evidence.

```
cp /etc/resolv.conf /root/resolv.backup && echo 'nameserver 192.0.2.1' >
/etc/resolv.conf && (dig +short +time=2 +tries=1 google.com || echo 'STEP 1:
resolution FAILS - symptom confirmed') && cp /root/resolv.backup /etc/resolv.conf &&
dig +short google.com
```

Test it — verification

Your record contains all six steps in the correct order, each with written evidence; a rollback position is stated before implementation; a specific preventive measure is named; and the DNS exercise shows the fault reproduced and then resolved.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the <code>apt-get install</code> step at the start of the lab — the playground starts with a minimal package set.
The KillerCoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against <code>labs/tools.md</code> . All four tools run entirely client-side and need no login.

Symptom	What to check
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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